

SQL Analysis & Query Results

Pizza Sales Analytics & Business Intelligence

End-to-End SQL Analysis using PostgreSQL

Prepared By

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Tools Used

- PostgreSQL
 - SQL
 - Microsoft Excel
 - Power BI (for dashboard validation)
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Project Description

This document contains the SQL queries executed during the Pizza Sales Analytics project along with their corresponding outputs and business interpretations.

The analysis was performed to validate data quality, calculate key performance indicators (KPIs), analyze sales trends, evaluate product performance, and uncover actionable business insights that support data-driven decision-making.

Analysis Sections

- Data Validation
- Key Performance Indicators (KPIs)
- Sales Trend Analysis
- Product Performance Analysis
- Advanced Business Insights

Project Type: Data Analytics & Business Intelligence

Database: PostgreSQL

KEY PERFORMANCE INDICATORS (KPIs)

KPI 1: Total Revenue

```
SELECT  
  ROUND(SUM(total_price), 2) AS total_revenue  
FROM pizza_sales;
```

	total_revenue numeric
1	817621.80

KPI 2: Total Orders

```
SELECT  
  COUNT(DISTINCT order_id) AS total_orders  
FROM pizza_sales;
```

	total_orders bigint
1	21334

KPI 3: Average Order Value

```
SELECT  
  ROUND(  
    SUM(total_price) / COUNT(DISTINCT order_id),  
    2  
  ) AS average_order_value  
FROM pizza_sales;
```

	average_order_value 
1	38.32

KPI 4: Total Pizzas Sold

```
SELECT
  SUM(quantity) AS total_pizzas_sold
FROM pizza_sales;
```

	total_pizzas_sold 
1	49559

KPI 5: Average Pizzas Per Order

```
SELECT
  ROUND(
    SUM(quantity)::NUMERIC /
    COUNT(DISTINCT order_id),
    2
  ) AS average_pizzas_per_order
FROM pizza_sales;
```

	average_pizzas_per_order 
1	2.32

SALES TREND ANALYSIS

1. Monthly Revenue Trend

```
SELECT
  EXTRACT(MONTH FROM order_date) AS month_number,
  TO_CHAR(order_date, 'Mon') AS month,
  ROUND(SUM(total_price), 2) AS total_revenue
FROM pizza_sales
GROUP BY
  month_number,
```

```
month
ORDER BY
month_number;
```

	month_number numeric	month text	total_revenue numeric
1	1	Jan	69776.55
2	2	Feb	65159.60
3	3	Mar	70397.10
4	4	Apr	68736.80
5	5	May	71238.25
6	6	Jun	68209.95
7	7	Jul	72557.90
8	8	Aug	68262.25
9	9	Sep	64180.05
10	10	Oct	64027.60
11	11	Nov	70374.60
12	12	Dec	64701.15

2. Daily Order Trend

```
SELECT
  EXTRACT(DOW FROM order_date) AS day_number,
  TRIM(TO_CHAR(order_date, 'Day')) AS day_name,
  COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY
  day_number,
  day_name
ORDER BY
  day_number;
```

	day_number numeric	day_name text	total_orders bigint
1	0	Sunday	2793
2	1	Monday	2972
3	2	Tuesday	3024
4	3	Wednesday	3238
5	4	Thursday	3535
6	5	Friday	3158
7	6	Saturday	2624

3. Percentage of Sales by Pizza Category

```

SELECT
  pizza_category,
  ROUND(
    SUM(total_price) * 100.0 /
    (SELECT SUM(total_price) FROM pizza_sales),
    2
  ) AS sales_percentage
FROM pizza_sales
GROUP BY
  pizza_category
ORDER BY
  sales_percentage DESC;

```

	pizza_category character varying (50)	sales_percentage numeric
1	Classic	26.91
2	Supreme	25.46
3	Chicken	23.95
4	Veggie	23.68

4. Percentage of Sales by Pizza Size

```

SELECT
  pizza_size,
  ROUND(
    SUM(total_price) * 100.0 /
    (SELECT SUM(total_price) FROM pizza_sales),
    2
  ) AS sales_percentage
FROM pizza_sales
GROUP BY
  pizza_size
ORDER BY
  sales_percentage DESC;

```

```

2
) AS sales_percentage
FROM pizza_sales
GROUP BY
    pizza_size
ORDER BY
    sales_percentage DESC;

```

	pizza_size character varying (10) 🔒	sales_percentage numeric 🔒
1	L	45.89
2	M	30.49
3	S	21.77
4	XL	1.72
5	XXL	0.12

5. Total Pizzas Sold by Pizza Size

```

SELECT
    pizza_size,
    SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY
    pizza_size
ORDER BY
    total_pizzas_sold DESC;

```

	pizza_size character varying (10) 🔒	total_pizzas_sold bigint 🔒
1	L	18951
2	M	15632
3	S	14396
4	XL	552
5	XXL	28

6. Total Pizzas Sold by Pizza Category

```

SELECT
    pizza_category,

```

```

SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY
    pizza_category
ORDER BY
    total_pizzas_sold DESC;

```

	pizza_category character varying (50) 🔒	total_pizzas_sold bigint 🔒
1	Classic	14884
2	Supreme	11983
3	Veggie	11646
4	Chicken	11046

PRODUCT PERFORMANCE ANALYSIS

1. Top 5 Pizzas by Revenue

```

SELECT
    pizza_name,
    ROUND(SUM(total_price), 2) AS total_revenue
FROM pizza_sales
GROUP BY
    pizza_name
ORDER BY
    total_revenue DESC
LIMIT 5;

```

	pizza_name character varying (100) 🔒	total_revenue numeric 🔒
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42730.50
3	The California Chicken Piz...	41388.75
4	The Classic Deluxe Pizza	38164.50
5	The Spicy Italian Pizza	34818.75

2. Bottom 5 Pizzas by Revenue

```
SELECT
    pizza_name,
    ROUND(SUM(total_price), 2) AS total_revenue
FROM pizza_sales
GROUP BY
    pizza_name
ORDER BY
    total_revenue ASC
LIMIT 5;
```

	pizza_name character varying (100) 	total_revenue numeric 
1	The Brie Carre Pizza	11588.50
2	The Green Garden Pizza	13943.75
3	The Spinach Supreme Piz...	15277.75
4	The Mediterranean Pizza	15360.50
5	The Spinach Pesto Pizza	15562.75

3. Top 5 Pizzas by Total Orders

```
SELECT
    pizza_name,
    COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY
    pizza_name
ORDER BY
    total_orders DESC
LIMIT 5;
```

	pizza_name character varying (100) 	total_orders bigint 
1	The Classic Deluxe Pizza	2328
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Piz...	2271
5	The Thai Chicken Pizza	2225

4. Bottom 5 Pizzas by Total Orders

```
SELECT
  pizza_name,
  COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY
  pizza_name
ORDER BY
  total_orders ASC
LIMIT 5;
```

	pizza_name character varying (100) 	total_orders bigint 
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Calabrese Pizza	918
4	The Spinach Supreme Piz...	918
5	The Chicken Pesto Pizza	938

5. Top 5 Pizzas by Quantity Sold

```
SELECT
  pizza_name,
  SUM(quantity) AS total_quantity_sold
FROM pizza_sales
GROUP BY
  pizza_name
ORDER BY
  total_quantity_sold DESC
LIMIT 5;
```

	pizza_name character varying (100) 	total_quantity_sold bigint 
1	The Classic Deluxe Pizza	2452
2	The Barbecue Chicken Piz...	2430
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

6. Bottom 5 Pizzas by Quantity Sold

```
SELECT
    pizza_name,
    SUM(quantity) AS total_quantity_sold
FROM pizza_sales
GROUP BY
    pizza_name
ORDER BY
    total_quantity_sold ASC
LIMIT 5;
```

	pizza_name character varying (100) 	total_quantity_sold bigint 
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Piz...	950
5	The Soppressata Pizza	960

ADVANCED BUSINESS ANALYSIS

1. Peak Sales Hours

```
SELECT
    EXTRACT(HOUR FROM order_time) AS hour_of_day,
    COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY
    hour_of_day
ORDER BY
    total_orders DESC;
```

	hour_of_day 	total_orders 
	numeric	bigint
1	12	2520
2	13	2455
3	18	2399
4	17	2335
5	19	2009
6	16	1920
7	20	1639
8	14	1472
9	15	1468
10	11	1231
11	21	1198
12	22	663
13	23	28
14	10	8
15	9	1

2. Weekday vs Weekend Performance

```

SELECT
  CASE
    WHEN EXTRACT(DOW FROM order_date) IN (0,6)
      THEN 'Weekend'
    ELSE 'Weekday'
  END AS day_type,
  COUNT(DISTINCT order_id) AS total_orders,
  ROUND(SUM(total_price),2) AS total_revenue
FROM pizza_sales
GROUP BY
  day_type;

```

	day_type text	total_orders bigint	total_revenue numeric
1	Weekday	15917	611104.75
2	Weekend	5417	206517.05

3. Average Revenue per Order

```

SELECT
  ROUND(
    AVG(order_total),
    2
  ) AS average_revenue_per_order
FROM
  (
    SELECT
      order_id,
      SUM(total_price) AS order_total
    FROM pizza_sales
    GROUP BY order_id
  ) AS orders;

```

	average_revenue_per_order numeric
1	38.32

4. Highest Revenue Generating Pizza Category

```

SELECT
  pizza_category,
  ROUND(SUM(total_price),2) AS total_revenue
FROM pizza_sales
GROUP BY
  pizza_category
ORDER BY
  total_revenue DESC;

```

	pizza_category character varying (50) 🔒	total_revenue numeric 🔒
1	Classic	219997.10
2	Supreme	208131.00
3	Chicken	195848.50
4	Veggie	193645.20

5. Best Selling Pizza Size

```
SELECT
    pizza_size,
    SUM(quantity) AS total_quantity_sold
FROM pizza_sales
GROUP BY
    pizza_size
ORDER BY
    total_quantity_sold DESC;
```

	pizza_size character varying (10) 🔒	total_quantity_sold bigint 🔒
1	L	18951
2	M	15632
3	S	14396
4	XL	552
5	XXL	28

6. Monthly Revenue Growth

```
SELECT
    EXTRACT(MONTH FROM order_date) AS month_number,
    TO_CHAR(order_date,'Mon') AS month,
    ROUND(SUM(total_price),2) AS total_revenue
FROM pizza_sales
GROUP BY
    month_number,
    month
ORDER BY
    month_number;
```

	month_number numeric	month text	total_revenue numeric
1	1	Jan	69776.55
2	2	Feb	65159.60
3	3	Mar	70397.10
4	4	Apr	68736.80
5	5	May	71238.25
6	6	Jun	68209.95
7	7	Jul	72557.90
8	8	Aug	68262.25
9	9	Sep	64180.05
10	10	Oct	64027.60
11	11	Nov	70374.60
12	12	Dec	64701.15

7. Revenue Contribution by Pizza Category

```

SELECT
  pizza_category,
  ROUND(
    SUM(total_price),
    2
  ) AS revenue,
  ROUND(
    SUM(total_price) * 100.0 /
    (
      SELECT SUM(total_price)
      FROM pizza_sales
    ),
    2
  ) AS revenue_percentage
FROM pizza_sales
GROUP BY
  pizza_category
ORDER BY
  revenue DESC;

```

	pizza_category character varying (50) 🔒	revenue numeric 🔒	revenue_percentage numeric 🔒
1	Classic	219997.10	26.91
2	Supreme	208131.00	25.46
3	Chicken	195848.50	23.95
4	Veggie	193645.20	23.68

8. Top 3 Pizzas Within Each Category

```

SELECT
    pizza_category,
    pizza_name,
    total_revenue,
    category_rank
FROM
(
    SELECT
        pizza_category,
        pizza_name,
        ROUND(SUM(total_price),2) AS total_revenue,
        DENSE_RANK() OVER
        (
            PARTITION BY pizza_category
            ORDER BY SUM(total_price) DESC
        ) AS category_rank
    FROM pizza_sales
    GROUP BY
        pizza_category,
        pizza_name
) ranked_pizzas
WHERE category_rank <= 3
ORDER BY
    pizza_category,
    category_rank;

```

	pizza_category character varying (50) 🔒	pizza_name character varying (100) 🔒	total_revenue numeric 🔒	category_rank bigint 🔒
1	Chicken	The Thai Chicken Pizza	43434.25	1
2	Chicken	The Barbecue Chicken Piz...	42730.50	2
3	Chicken	The California Chicken Piz...	41388.75	3
4	Classic	The Classic Deluxe Pizza	38164.50	1
5	Classic	The Hawaiian Pizza	32273.25	2
6	Classic	The Pepperoni Pizza	30161.75	3
7	Supreme	The Spicy Italian Pizza	34818.75	1
8	Supreme	The Italian Supreme Pizza	33476.75	2
9	Supreme	The Sicilian Pizza	30920.25	3
10	Veggie	The Four Cheese Pizza	32265.70	1
11	Veggie	The Mexicana Pizza	26780.75	2
12	Veggie	The Five Cheese Pizza	26066.50	3

9. Running Monthly Revenue (Window Function)

```

SELECT
    month,
    total_revenue,
    SUM(total_revenue) OVER
    (
        ORDER BY month_number
    ) AS cumulative_revenue
FROM
    (
        SELECT
            EXTRACT(MONTH FROM order_date) AS month_number,
            TO_CHAR(order_date,'Mon') AS month,
            ROUND(SUM(total_price),2) AS total_revenue
        FROM pizza_sales
        GROUP BY
            month_number,
            month
    ) monthly_sales
ORDER BY
    month_number;
```


	month text	total_revenue numeric	cumulative_revenue numeric
1	Jan	69776.55	69776.55
2	Feb	65159.60	134936.15
3	Mar	70397.10	205333.25
4	Apr	68736.80	274070.05
5	May	71238.25	345308.30
6	Jun	68209.95	413518.25
7	Jul	72557.90	486076.15
8	Aug	68262.25	554338.40
9	Sep	64180.05	618518.45
10	Oct	64027.60	682546.05
11	Nov	70374.60	752920.65
12	Dec	64701.15	817621.80

10. Top 10 Highest Revenue Orders

```

SELECT
  order_id,
  ROUND(SUM(total_price),2) AS order_revenue
FROM pizza_sales
GROUP BY
  order_id
ORDER BY
  order_revenue DESC
LIMIT 10;

```

	order_id integer 	order_revenue numeric 
1	18845	444.20
2	10760	417.15
3	1096	285.15
4	6169	284.00
5	740	280.95
6	12257	276.75
7	1685	272.75
8	4482	267.20
9	5452	267.10
10	16786	265.55